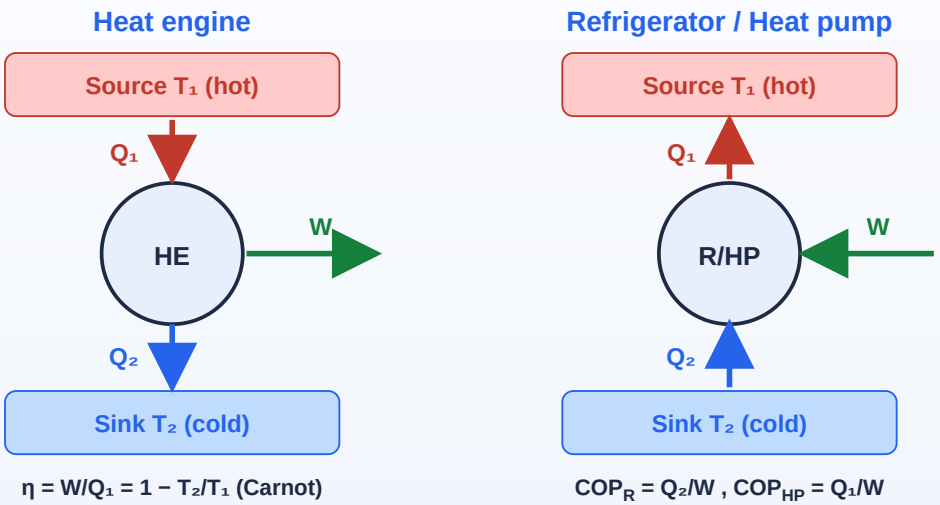
UPSSSC ALERT

UPSSSC JE

MECHANICAL THERMODYNAMICS

Complete English Edition

Unit 4 · 5 Marks · 2026



$\eta = W/Q_1 = 1 - T_2/T_1$ (Carnot)

$COP_R = Q_2/W$, $COP_{HP} = Q_1/W$

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UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Thermodynamics**. Syllabus: properties of pure substances, steam tables, steam formation, wet and superheated steam, dryness fraction, degree of superheat, first law, steady flow process, SFEE, second law, heat source, sink and reservoir, heat engine, heat pump and refrigerator.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (P. K. Nag, R. K. Rajput, R. S. Khurmi).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Pure Substances & Properties of Steam

Pure Substances & Properties of Steam – Basic Concept

A **pure substance** has a uniform chemical composition in all phases — e.g. a mixture of water, ice and steam. Steam is the working fluid of power plants, so steam formation, its states (wet, dry saturated, superheated) and the use of **steam tables** must be understood.

Understand in Detail — Pure Substances & Properties of Steam

1.1 Pure substance and the story of steam formation

A **pure substance** is one whose chemical composition is the same everywhere and in every phase — whether solid, liquid, gas or a mixture of these. A mixture of water, ice and steam is a pure substance (H_2O everywhere). But a mixture of air and liquid air is not (on liquefaction the nitrogen–oxygen ratio changes).

Steam formation at constant pressure — an experiment: keep 1 kg of water at 25°C in a cylinder with a weighted piston on top (pressure fixed at 1 atm). (1) Add heat — the water temperature rises, volume nearly the same; this is **subcooled (compressed) liquid**. Heat in this stage = **sensible heat (h_f)**. (2) On reaching 100°C the water is **saturated liquid** — now adding more heat does **not** raise the temperature, water keeps turning into steam; this heat = **latent heat (h_{fg})**, 2257 kJ/kg at 1 atm. The in-between state = **wet steam**. (3) When the last drop has become vapour — **dry saturated steam**. (4) Adding more heat raises the steam temperature again — **superheated steam**; this heat = $c_{ps}(T_{sup} - T_s)$.

What changes when pressure increases? At higher pressure water boils at a higher temperature (≈ 120°C in a pressure cooker), and latent heat **decreases**. At one pressure — the **critical point (221.2 bar, 374.15°C)** — latent heat becomes zero; water turns into steam directly without boiling or bubbles. Pressures above this are **supercritical**. At the bottom is the **triple point (0.01°C, 0.006113 bar)** — where solid, liquid and vapour coexist in equilibrium.

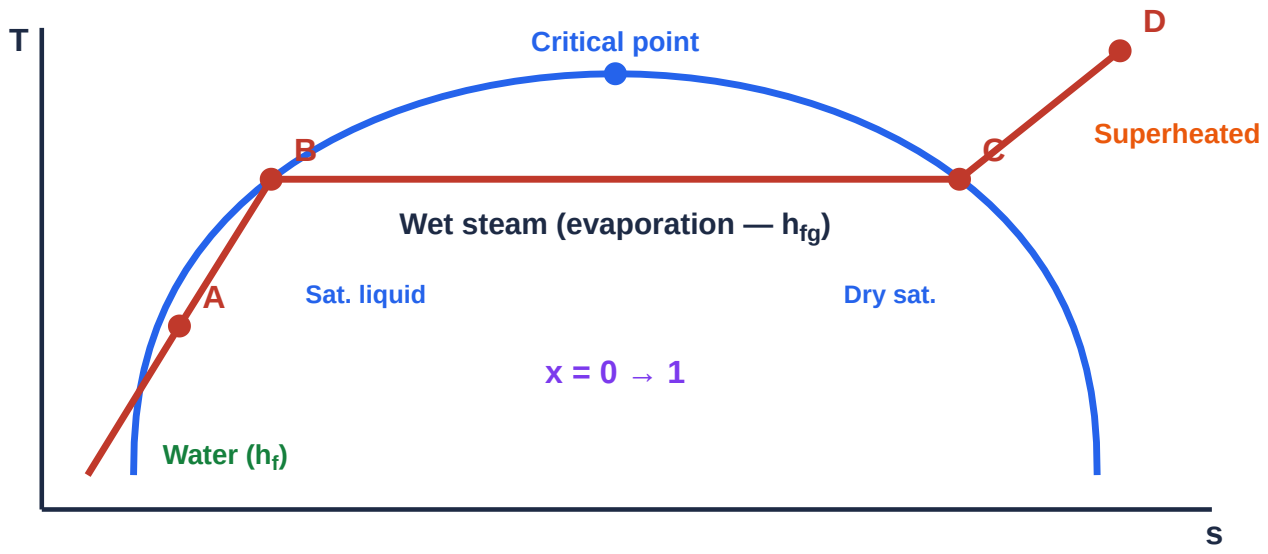


Fig 4.1: Steam formation on T–s diagram (A–B water, B–C evaporation, C–D superheating)

1.2 Dryness fraction and calculating steam properties

How much vapour and how much water there is in wet steam is told by the **dryness fraction (x)**: **x = mass of dry steam ÷ (mass of dry steam + mass of water droplets)**. For dry saturated steam x = 1, for saturated water x = 0. **(1 − x) = wetness fraction**.

Properties of wet steam — think that in wet steam a fraction x has fully become vapour and (1 − x) is still water. So only the fraction x of latent heat has been supplied:

Derivation: Formulae for wet and superheated steam

1. **Enthalpy:** $h = h_f + x\, h_{fg}$
2. **Specific volume:** $v = v_f + x\, v_{fg} \approx x\, v_g$ (water volume negligible)
3. **Entropy:** $s = s_f + x\, s_{fg}$; and $s_{fg} = h_{fg}/T_s$
4. **Superheated steam:** $h_{sup} = h_g + c_{ps}(T_{sup} - T_s)$; $c_{ps} \approx 2.1$ kJ/kg·K ; **degree of superheat** = $T_{sup} - T_s$
5. **Internal energy:** $u = h - pv$

Instruments to measure dryness fraction (calorimeters): (1) **barrel** — condensing steam in cold water; (2) **throttling calorimeter** — steam is throttled to make it superheated; in throttling h is constant, so $h_{f1} + x\, h_{fg1} = h_{sup2}$ → x is found; works only for **nearly dry** steam (x > 0.95); (3) **separating** — droplets separated mechanically, for very wet steam; (4) **combined separating-throttling** — for any steam, most accurate: $x = x_1 \times x_2$.

Example: Find the enthalpy of steam at 10 bar with dryness 0.9. (h_f = 762.6, h_{fg} = 2015.3 kJ/kg, T_s = 179.9°C)

Solution:

1. h = 762.6 + 0.9 × 2015.3 = 762.6 + 1813.8 = **2576.4 kJ/kg**
2. If the same steam is superheated to 250°C: h = 2778.0 + 2.1 × (250 − 179.9) = **2925.2 kJ/kg** (h_g = 762.6 + 2015.3 ≈ 2778)

1.3 Using steam tables and the Mollier (h–s) chart

Steam tables are of two kinds: (1) **pressure-based** — for each pressure the saturation temperature, v_g, h_f, h_{fg}, h_g, s_f, s_{fg}, s_g; (2) **temperature-based** — for each temperature the saturation pressure and the same properties. Superheated steam has a separate table read with both pressure and temperature.

Mollier chart (h–s diagram): enthalpy on the vertical axis, entropy on the horizontal. Its biggest advantage — **isentropic expansion in a turbine or nozzle is a straight vertical line**, and the heat drop ($h_1 - h_2$) can be read directly from its length. Below the saturation line are lines of constant dryness (x), above it lines of constant temperature; constant-pressure lines run through both regions.

Note: Area on a T–s diagram = heat (since dQ = T ds); area on a p–v diagram = work. On a T–s diagram, inside the saturation dome a constant-pressure line is **horizontal** (temperature constant during evaporation).

1. Key Points

- **Steam formation (at constant pressure):** (i) heating water to saturation temperature — sensible heat h_f; (ii) evaporation — latent heat h_{fg} (temperature constant); (iii) superheating — $c_{ps}(T_{sup} - T_s)$.
- **Saturation temperature:** the boiling temperature at a given pressure; it rises with pressure (100°C at 1.013 bar).
- **Wet steam:** steam carrying water droplets; dryness fraction x = mass of dry steam/total mass. $h = h_f + x\,h_{fg}$, $v = x\,v_g$.
- **Dry saturated steam:** x = 1; $h = h_g = h_f + h_{fg}$. **Superheated steam:** steam heated above saturation temperature; $h_{sup} = h_g + c_{ps}(T_{sup} - T_s)$.
- **Degree of superheat:** $T_{sup} - T_s$. Superheated steam behaves nearly like an ideal gas; $c_{ps} \approx 2.1$ kJ/kg·K.
- **Critical point:** 220.9 bar and 374.15°C; here $h_{fg} = 0$ — water flashes directly into steam. **Triple point:** 0.006113 bar, 0.01°C — all three phases coexist.
- **Steam tables:** saturated steam tables (pressure- or temperature-based: T_s, v_g, h_f, h_{fg}, h_g, s_f, s_{fg}, s_g) and superheated steam tables. Reference: u and s of saturated water at 0.01°C are zero.
- **Calorimeters for dryness fraction:** bucket, throttling, separating, and combined separating-throttling calorimeter (for very wet steam).

- **Mollier (h–s) chart:** used to read enthalpy drops directly, especially in isentropic expansion of steam.
- **Other pure substances:** water–steam, air (in gas phase, homogeneous mixture), ammonia. An oil–water mixture is not a pure substance.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Enthalpy of wet steam	$h = h_f + x\, h_{fg}$
Dry saturated steam	$h_g = h_f + h_{fg}$
Superheated steam	$h = h_g + c_{ps}(T_{sup} - T_s)$
Specific volume of wet steam	$v = x\, v_g$
Entropy of wet steam	$s = s_f + x\, s_{fg}$
Internal energy	$u = h - pv$
Dryness fraction	$x = m_g / (m_g + m_f)$
Throttling calorimeter	$h_{f1} + x\, h_{fg1} = h_{g2} + c_{ps}(T_{sup} - T_{s2})$
Combined calorimeter	$x = x_1 \times x_2$
Degree of superheat	$T_{sup} - T_s$

3. Detailed Explanation

- **T–s and p–v diagrams:** the saturated liquid line (x = 0) and saturated vapour line (x = 1) meet at the critical point to form a dome; inside is wet steam, left is compressed liquid, right is superheated steam.
- **Effect of increasing pressure:** T_s and h_f increase, h_{fg} decreases to zero at the critical point; h_g rises up to about 30 bar and then falls.
- **Principle of throttling calorimeter:** throttling wet steam to low pressure (h constant) makes it superheated; measuring temperature gives h_2, and $h_1 = h_2$ gives x. Steam must not be too wet ($x > \approx 0.93$).
- **Advantages of superheated steam:** higher heat content, less condensation and erosion in turbines, higher efficiency; hence modern plants use superheated steam.
- **External work of evaporation:** $p(v_g - v_f) \approx p\, v_g$; internal latent heat = $h_{fg} - p\, v_g$.

4. Comparison Table — Wet, Dry Saturated and Superheated Steam

Basis	Wet	Dry saturated	Superheated
Dryness fraction	$x < 1$	$x = 1$	Not applicable
Temperature	T_s	T_s	$T > T_s$
Enthalpy	$h_f + x\, h_{fg}$	h_g	$h_g + c_{ps}\Delta T$
Behaviour	With water droplets	Invisible pure vapour	Like an ideal gas

5. Solved Numerical Problems

Example 1. Steam at 10 bar has dryness 0.9 ($h_f = 762.6$, $h_{fg} = 2015.3$ kJ/kg). Find the enthalpy.
Solution:

1. $h = 762.6 + 0.9 \times 2015.3 = 762.6 + 1813.8 =$ **2576.4 kJ/kg**

Example 2. Steam at 10 bar ($T_s = 179.9$°C, $h_g = 2777.1$) is superheated to 250°C; $c_{ps} = 2.1$. Find degree of superheat and enthalpy.
Solution:

1. Degree of superheat $= 250 - 179.9 =$ **70.1°C**
2. $h = 2777.1 + 2.1 \times 70.1 =$ **2924.3 kJ/kg**

Example 3. 2 kg of wet steam contains 0.2 kg of water. Find the dryness fraction.
Solution:

1. $x = 1.8/2 =$ **0.9**

Example 4. Steam at 5 bar ($v_g = 0.3749$ m³/kg) has $x = 0.8$. Find its specific volume.
Solution:

1. $v = x\, v_g = 0.8 \times 0.3749 =$ **0.300 m³/kg**

Example 5. In a separating-throttling calorimeter $x_1 = 0.95$ and $x_2 = 0.97$. Find the actual dryness fraction.
Solution:

1. $x = x_1 \times x_2 = 0.95 \times 0.97 =$ **0.9215**

6. Frequently Asked Facts

- Critical point: 220.9 bar, 374.15°C; latent heat is zero there.
- Wet steam $h = h_f + x\, h_{fg}$; superheated $h = h_g + c_{ps}(T_{sup} - T_s)$.
- Combined separating-throttling calorimeter for very wet steam.
- Latent heat decreases as pressure increases.

7. Note (Confusing Facts)

- For dry saturated steam $x = 1$, but x is not defined for superheated steam (it does not exceed 1).

- Temperature stays constant during evaporation even though heat is supplied (latent heat).
- The triple point temperature is 0.01°C, not 0°C.

Remember: One line: x = 0 saturated water, x = 1 dry saturated steam. Assertion–Reason: (A) Superheated steam is used in turbines. (R) It reduces blade erosion and increases efficiency. — Both true; R explains A.

30 Most Important One-Liner Facts — Pure Substances & Properties of Steam

1. Pure substance?

— Uniform chemical composition in all phases.

2. Saturation temperature at 1.013 bar?

— 100°C.

3. Symbol of sensible heat?

— h_f.

4. Symbol of latent heat?

— h_{fg}.

5. Temperature during evaporation?

— Constant.

6. Dryness fraction?

— Mass of dry steam ÷ total mass.

7. x for saturated water?

— 0.

8. x for dry saturated steam?

— 1.

9. Enthalpy of wet steam?

— $h_f + x\ h_{fg}$.

10. Enthalpy of superheated steam?

— $h_g + c_{ps}(T_{sup} - T_s)$.

11. Degree of superheat?

— $T_{sup} - T_s$.

12. c_{ps} of superheated steam?

— ≈ 2.1 kJ/kg·K.

13. Critical pressure?

— 220.9 bar.

14. Critical temperature?

— 374.15°C.

15. h_{fg} at critical point?

— Zero.

16. Triple point temperature of water?

— 0.01°C.

17. Specific volume of wet steam?

— $x\ v_g$.

18. Entropy of wet steam?

— $s_f + x\ s_{fg}$.

19. h_{fg} with rising pressure?

— Decreases.

20. T_s with rising pressure?

— Increases.

21. Calorimeter for very wet steam?

— Combined separating-throttling.

22. Principle of throttling calorimeter?

— $h_1 = h_2$.

23. x in combined calorimeter?

— $x_1 \times x_2$.

24. Name of h–s chart?

— Mollier diagram.

25. Reference state in steam tables?

— Saturated water at 0.01°C.

26. State inside the dome?

— Wet steam.

27. Right of the dome?

— Superheated steam.

28. External work of evaporation?

— $p\ v_g$ (approx.).

29. Superheated steam behaves like?

— Ideal gas.

30. Internal energy (steam)?

— u = h − pv.

Practice MCQs — Pure Substances & Properties of Steam

Q1. Enthalpy of wet steam:

(a) $h_f + h_{fg}$

(b) $h_f + x\ h_{fg}$

(c) $x\ h_f + h_{fg}$

(d) $h_g + x\ h_{fg}$

✔ **Answer: (b) $h_f + x\ h_{fg}$**

Explanation: Fraction x is evaporated.

Q2. Latent heat at the critical point:

(a) Maximum

(b) Zero

(c) 2257 kJ/kg

(d) Infinite

✔ **Answer: (b) Zero**

Explanation: Liquid–vapour distinction vanishes.

Q3. Dryness fraction of dry saturated steam:

(a) 0

(b) 0.5

(c) 1

(d) More than 1

✔ **Answer: (c) 1**

Explanation: x = 1.

Q4. Critical pressure of water:

(a) 100 bar

(b) 220.9 bar

(c) 1.013 bar

(d) 374 bar

✔ **Answer: (b) 220.9 bar**

Explanation: 220.9 bar.

Q5. Latent heat of evaporation as pressure increases:

(a) Increases

(b) Decreases

(c) Remains constant

(d) First decreases then increases

✔ **Answer: (b) Decreases**

Explanation: Zero at the critical point.

Q6. To measure dryness of very wet steam we use:

(a) Throttling calorimeter

(b) Combined separating-throttling calorimeter

(c) Bomb calorimeter

(d) Junker's calorimeter

✔ **Answer: (b) Combined separating-throttling calorimeter**

Explanation: Throttling alone cannot superheat it.

Q7. Triple point temperature of water:

(a) 0°C
(b) 0.01°C
(c) 4°C
(d) 100°C

✔ **Answer: (b) 0.01°C**

Explanation: 0.01°C, 0.006113 bar.

Q8. Enthalpy of superheated steam:

(a) $h_g + c_{ps}(T_{sup} - T_s)$
(b) $h_f + c_{ps}T$
(c) $h_{fg} + c_{ps}\Delta T$
(d) $x\ h_g$

✔ **Answer: (a) $h_g + c_{ps}(T_{sup} - T_s)$**

Explanation: Superheat added.

Q9. 2 kg wet steam with 0.4 kg water — dryness fraction:

(a) 0.2
(b) 0.6
(c) 0.8
(d) 0.4

✔ **Answer: (c) 0.8**

Explanation: 1.6/2.

Q10. Temperature during evaporation:

(a) Increases
(b) Decreases
(c) Remains constant
(d) Uncertain

✔ **Answer: (c) Remains constant**

Explanation: Latent heat.

Q11. The h–s chart is called:

(a) Mollier diagram
(b) T–s diagram
(c) Psychrometric chart
(d) p–v diagram

✔ **Answer: (a) Mollier diagram**

Explanation: Mollier.

Q12. Specific volume of wet steam is approximately:

(a) v_f
(b) $x\ v_g$
(c) v_g/x
(d) $v_g + v_f$

✔ **Answer: (b) $x\ v_g$**

Explanation: v_f is negligible.

Q13. Throttling calorimeter works on:

(a) Constant entropy
(b) Constant enthalpy
(c) Constant temperature
(d) Constant volume

✔ **Answer: (b) Constant enthalpy**

Explanation: $h_1 = h_2$.

Q14. At 10 bar T_s = 180°C, steam at 230°C — degree of superheat:

(a) 50°C
(b) 230°C
(c) 180°C
(d) 410°C

✔ **Answer: (a) 50°C**

Explanation: 230 − 180.

Q15. Which is not a pure substance?

(a) Water and steam
(b) Ice and water
(c) Oil and water
(d) Ammonia

✔ **Answer: (c) Oil and water**

Explanation: Heterogeneous, different composition.

SSC JE / UPSSSC JE Previous Year Questions — Pure Substances & Properties of Steam

Q1. Dryness fraction of steam is defined as: SSC JE (Previous Year)

(a) Mass of dry steam/total mass
(b) Mass of water/total mass
(c) Dry steam/water
(d) Water/dry steam

✔ **Answer: (a) Mass of dry steam/total mass**

Explanation: Definition.

Q2. Temperature of water at the critical point: SSC JE (Previous Year)

(a) 100°C
(b) 273°C
(c) 374.15°C
(d) 500°C

✔ **Answer: (c) 374.15°C**

Explanation: 374.15°C.

Q3. Temperature of superheated steam is: SSC JE (Previous Year)

(a) Below saturation temperature
(b) Equal to saturation temperature
(c) Above saturation temperature
(d) 0°C

✔ **Answer: (c) Above saturation temperature**

Explanation: Definition.

Q4. In throttling of steam, ____ remains constant: SSC JE (Previous Year)

(a) Enthalpy
(b) Entropy
(c) Pressure
(d) Temperature

✔ **Answer: (a) Enthalpy**

Explanation: $h_1 = h_2$.

Q5. The point where all three phases coexist: SSC JE (Previous Year)

(a) Critical point

(b) Triple point

(c) Boiling point

(d) Freezing point

✔ **Answer: (b) Triple point**

Explanation: Triple point.

Q6. Enthalpy of dry saturated steam: SSC JE (Previous Year)

(a) h_f

(b) h_{fg}

(c) $h_f + h_{fg}$

(d) $h_f - h_{fg}$

✔ **Answer: (c) $h_f + h_{fg}$**

Explanation: h_g.

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Chapter 2: Basic Concepts & First Law of Thermodynamics

Basic Concepts & First Law of Thermodynamics – Basic Concept

Thermodynamics is the science of energy, heat and work, their inter-conversion and their relation to the properties of matter. The **first law** is the law of conservation of energy — energy can neither be created nor destroyed, only converted. Engines, turbines, compressors and nozzles are analysed with it.

Understand in Detail — Basic Concepts & First Law of Thermodynamics

2.1 The language of thermodynamics — system, property, state, process

Thermodynamics is the science of energy, especially the inter-conversion of heat and work. To understand it, a few words must first be clear.

System — the body or space we are studying; everything outside it is the **surroundings**; the surface separating them is the **boundary**, which may be fixed or moving. **Closed system** — no mass crosses, only energy (gas in a piston-cylinder). **Open system (control volume)** — both mass and energy (turbine, pump, nozzle, compressor). **Isolated** — nothing crosses (the universe, an ideal thermos flask).

Property — a characteristic of a system that depends only on its present state, not on the path taken to reach it (pressure, temperature, volume, internal energy, enthalpy, entropy). **Intensive** — independent of mass (p, T, ρ, specific properties); **extensive** — depends on mass (V, U, H, mass). **Heat and work are not properties** — they are **path functions**, i.e. their value depends on the path; so their differentials are inexact (δQ, δW).

Process — a change from one state to another. **Cycle** — a process that returns to the initial state. **Quasi-static** — so slow that at every instant the system is close to equilibrium. **Zeroth law**: if A is in thermal equilibrium with C and B is also with C, then A and B are in thermal equilibrium with each other — **this is the basis of the thermometer**.

2.2 Work and heat — the two 'moving' forms of energy

Heat — energy crossing the boundary only due to a **temperature difference**. **Work** — energy crossing the boundary due to any cause other than temperature difference, whose sole external effect could be the raising of a weight. Both are 'energy in transit' — a system does not 'contain' heat; heat 'flows in or out'.

Sign convention: heat supplied to the system is **positive**, rejected is negative; work done by the system is **positive** (e.g. gas expanding and pushing a piston), work done on the system is negative (compression).

Derivation: Displacement work $W = \int p\,dV$ in different processes

1. **Constant volume (isochoric):** $dV = 0 \rightarrow \mathbf{W = 0}$; $Q = m\,c_v\,\Delta T = \Delta U$
2. **Constant pressure (isobaric):** $W = p(V_2 - V_1)$; $Q = m\,c_p\,\Delta T = \Delta H$
3. **Isothermal ($pV = C$):** $W = p_1V_1 \ln(V_2/V_1)$; $\Delta U = 0$ (ideal gas) $\rightarrow Q = W$
4. **Adiabatic ($pV^{\gamma} = C$):** $W = (p_1V_1 - p_2V_2)/(\gamma - 1)$; $Q = 0 \rightarrow W = -\Delta U$
5. **Polytropic ($pV^{n} = C$):** $W = (p_1V_1 - p_2V_2)/(n - 1)$; $\mathbf{Q = W\,(\gamma - n)/(\gamma - 1)}$
6. Identify by n: n = 0 constant pressure, n = 1 isothermal, n = γ adiabatic, n = ∞ constant volume

Ideal gas relations: $pV = mRT$; $c_p - c_v = R$; $\gamma = c_p/c_v$; for air $R = 0.287$ kJ/kg·K, $c_p = 1.005$, $c_v = 0.718$, $\gamma = 1.4$. **Universal gas constant** $\bar{R} = 8.314$ kJ/kmol·K = R × molecular weight. **Free expansion** — a gas expanding into vacuum: W = 0, Q = 0, so ΔU = 0 (temperature of an ideal gas does not change), but the process is irreversible.

2.3 First law — the accounting of energy

The **first law** is the law of conservation of energy: energy can neither be created nor destroyed, only converted from one form to another. **Joule's experiment**: water in a closed vessel was stirred with a paddle (work done) and it heated up; then it was allowed to cool (heat removed) — for the whole cycle $\oint \delta Q = \oint \delta W$. That is, **for a cycle, net heat = net work**.

For a process: if heat and work are not equal, the difference is stored inside the system — as **internal energy (U)**: $\mathbf{Q - W = \Delta U}$ (or $\delta Q = dU + \delta W$). Internal energy is the sum of molecular kinetic and potential energies; for an ideal gas it depends **only on temperature** (Joule's law): $\Delta U = m\,c_v\,\Delta T$ — whatever the process.

Enthalpy ($H = U + pV$) — a convenient property that appears naturally in flow systems, because a flowing fluid brings with it not only internal energy but also **flow work (pV)**. For an ideal gas $\Delta H = m\,c_p\,\Delta T$. A **perpetual motion machine of the first kind (PMM-1)** — one that gives continuous work without taking energy — is impossible by the first law.

> **Example:** 1 kg of air is heated at constant pressure from 1 bar, 27°C to 127°C. Find Q, ΔU and W.
> **Solution:**
> 1. $Q = m\,c_p\,\Delta T = 1 \times 1.005 \times 100 =$ **100.5 kJ**
> 2. $\Delta U = m\,c_v\,\Delta T = 1 \times 0.718 \times 100 =$ **71.8 kJ**
> 3. $W = Q - \Delta U =$ **28.7 kJ** (check: $W = mR\Delta T = 0.287 \times 100 = 28.7$ ✔)

2.4 Steady flow energy equation (SFEE) and its applications

In machines like turbines, compressors and nozzles the fluid flows continuously. If the flow rate and all properties do not change with time it is called **steady flow**. For such an open system the energy account is: **energy in = energy out**.

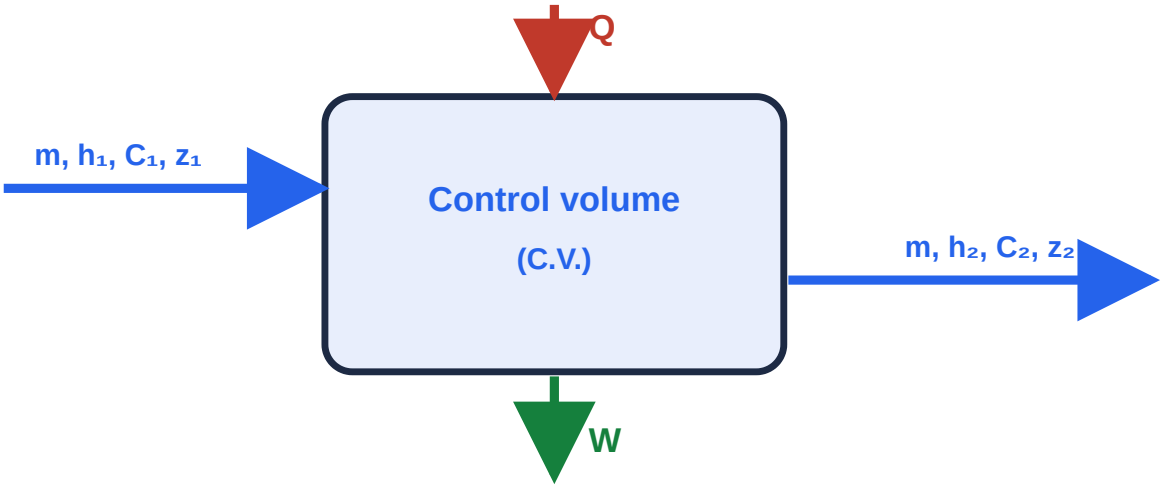


Fig 4.2: Steady Flow Energy Equation (SFEE)

Derivation: SFEE (per kg)

1. At inlet the fluid brings: $h_1 + V_1^2/2 + gz_1$; carries out: $h_2 + V_2^2/2 + gz_2$
2. $\mathbf{h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w}$
3. **Turbine** (adiabatic, negligible velocity and height change): $w = h_1 - h_2$
4. **Compressor/pump**: $w = -(h_2 - h_1)$ (work is supplied)
5. **Nozzle** ($w = 0$, $q = 0$): $V_2 = \sqrt{[2(h_1 - h_2) + V_1^2]}$
6. **Throttling valve** ($w = 0$, $q = 0$, negligible velocity change): $\mathbf{h_1 = h_2}$ — constant enthalpy ; **boiler/condenser** ($w = 0$): $q = h_2 - h_1$

Example: Steam enters a turbine with enthalpy 3000 kJ/kg and velocity 50 m/s, leaves with 2300 kJ/kg and 150 m/s; heat loss negligible. Work per kg?

Solution:

1. $\Delta(V^2/2) = (150^2 - 50^2)/2 = 10000$ J/kg = 10 kJ/kg
2. $w = (3000 - 2300) - 10 =$ **690 kJ/kg** — notice the kinetic energy contribution is small, which is why it is often neglected

1. Key Points

- **Types of system:** closed (energy but no mass transfer), open (both mass and energy) and isolated (neither). An open system is called a control volume.
- **Property:** intensive (independent of mass: pressure, temperature, density) and extensive (depends on mass: volume, energy, enthalpy). Extensive ÷ mass = specific property (intensive).
- **Thermodynamic equilibrium:** thermal (equal temperature), mechanical (equal pressure) and chemical equilibrium together.

- **Zeroth law:** if two systems are each in thermal equilibrium with a third, they are in equilibrium with each other — the basis of thermometry.
- **Heat and work:** both are boundary phenomena and path functions, not properties; their differentials are inexact. Sign convention: heat supplied to the system +, work done by the system +.
- **First law (closed system):** $Q = \Delta U + W$; for a cycle $\oint \delta Q = \oint \delta W$. Internal energy (U) is a property.
- **Enthalpy:** $H = U + pV$; for an ideal gas $h = c_p T$ and $u = c_v T$. $c_p - c_v = R$ and $\gamma = c_p/c_v$ (1.4 for air).
- **Perpetual motion machine of the first kind (PMM-I):** a machine producing work continuously without energy input is impossible — first law.
- **Steady flow process:** properties inside the control volume do not change with time; mass flow in = mass flow out.
- **SFEE:** $h_1 + C_1^2/2 + gz_1 + q = h_2 + C_2^2/2 + gz_2 + w$ (per kg); flow work pv is included in enthalpy.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
First law (closed)	$Q = \Delta U + W$
Cycle	$\oint \delta Q = \oint \delta W$
Enthalpy	$H = U + pV$
Ideal gas	$pV = mRT$, $c_p - c_v = R$
Constant volume work	$W = 0$, $Q = mc_v\Delta T$
Constant pressure work	$W = p(V_2 - V_1)$, $Q = mc_p\Delta T$
Isothermal work	$W = p_1V_1 \ln(V_2/V_1)$
Adiabatic ($pV^\gamma = C$)	$W = (p_1V_1 - p_2V_2)/(\gamma - 1)$
Polytropic ($pV^n = C$)	$W = (p_1V_1 - p_2V_2)/(n - 1)$
SFEE	$h_1 + C_1^2/2 + gz_1 + q = h_2 + C_2^2/2 + gz_2 + w$
Nozzle exit velocity	$C_2 = \sqrt{[2(h_1 - h_2)]}$ (C_1 negligible)

3. Detailed Explanation

- **Summary of processes:** constant volume (isochoric) – zero work; constant pressure (isobaric); isothermal – for ideal gas $\Delta U = 0$, $Q = W$; adiabatic – $Q = 0$; throttling – $h_1 = h_2$ (isenthalpic).
- **Applications of SFEE:** turbine ($q \approx 0$, ΔKE, ΔPE negligible): $w = h_1 - h_2$; compressor: $w = -(h_2 - h_1)$; nozzle ($q = w = 0$): $C_2 = \sqrt{(2\Delta h)}$; boiler/condenser ($w = 0$): $q = h_2 - h_1$; throttle valve: $h_1 = h_2$.
- **Quasi-static process:** an infinitely slow process with the system always near equilibrium; only such a process can be drawn as a line on a p–V diagram and its work is $\int p\, dV$.
- **Free expansion:** expansion into vacuum — $W = 0$, $Q = 0$, so $\Delta U = 0$; the temperature of an ideal gas stays the same, but the process is irreversible.

- **Joule's experiment:** paddle-wheel work equalled the heat produced, establishing the equivalence of heat and work (1 cal = 4.186 J).

4. Comparison Table — Closed vs Open System

Basis	Closed system	Open system
Mass transfer	No	Yes
Energy transfer	Yes	Yes
First law	$Q = \Delta U + W$	SFEE
Examples	Closed piston–cylinder	Turbine, compressor, nozzle

4. Comparison Table — Point vs Path Function

Basis	Point function (property)	Path function
Depends on	End states only	Path of process
Differential	Exact	Inexact
Examples	p, V, T, U, H, S	Heat, work

5. Solved Numerical Problems

Example 1. A closed system receives 100 kJ of heat and does 40 kJ of work. Find the change in internal energy.

Solution:

1. $\Delta U = Q - W = 100 - 40 =$ **60 kJ** (increase)

Example 2. 1 kg of air is compressed at a constant 1 bar from 0.8 m^3 to 0.5 m^3. Find the work.

Solution:

1. $W = p(V_2 - V_1) = 100 \times (0.5 - 0.8) =$ **−30 kJ**
2. Negative sign: work done on the system

Example 3. Steam enters a turbine at 3200 kJ/kg and leaves at 2600 kJ/kg; flow 2 kg/s. Find the power (neglect heat loss, KE, PE).

Solution:

1. $w = h_1 - h_2 = 600$ kJ/kg
2. $P = 2 \times 600 =$ **1200 kW**

Example 4. Enthalpy drop in a nozzle is 180 kJ/kg; inlet velocity negligible. Find the exit velocity.

Solution:

1. $C_2 = \sqrt{(2 \times 180 \times 1000)} = \sqrt{360000} =$ **600 m/s**

Example 5. 0.1 m^3 of air at 10 bar expands isothermally to 1 bar. Find the work.

Solution:

1. $W = p_1V_1 \ln(p_1/p_2) = 1000 \times 0.1 \times \ln 10$
2. $W = 100 \times 2.303 =$ **230.3 kJ**

6. Frequently Asked Facts

- Nozzle from SFEE: $C_2 = 44.72\sqrt{\Delta h}$ (Δh in kJ/kg).
- Enthalpy remains constant in throttling.
- Heat and work are path functions; internal energy and enthalpy are properties.
- Work is zero in a constant volume process.

7. Note (Confusing Facts)

- Work is zero in free expansion even though volume increases.
- In an isothermal process heat is supplied yet temperature does not change (Q = W for ideal gas).
- Pressure and temperature are intensive; volume and energy are extensive.

Remember: One line: closed system Q = ΔU + W; open system SFEE. Assertion–Reason: (A) PMM-I is impossible. (R) It violates conservation of energy. — Both true; R explains A.

30 Most Important One-Liner Facts — Basic Concepts & First Law of Thermodynamics

1. System with only energy transfer?

— Closed system.

2. Both mass and energy transfer?

— Open system.

3. No transfer at all?

— Isolated system.

4. Another name for open system?

— Control volume.

5. Property independent of mass?

— Intensive property.

6. Volume is which property?

— Extensive.

7. Basis of thermometry?

— Zeroth law.

8. Law of energy conservation?

— First law.

9. First law for closed system?

— Q = ΔU + W.

10. For a cyclic process?

— $\oint \delta Q = \oint \delta W$.

11. Heat and work are?

— Path functions.

12. Internal energy is?

— A property (point function).

13. Enthalpy?

— H = U + pV.

14. $c_p - c_v$ =

— R.

15. γ of air?

— 1.4.

16. R for air?

— 0.287 kJ/kg·K.

17. Work in constant volume process?

— Zero.

18. Heat in adiabatic process?

— Zero.

19. ΔU in isothermal (ideal gas)?

— Zero.

20. Constant in throttling?

— Enthalpy.

21. Work in free expansion?

— Zero.

22. PMM-I violates?

— First law.

23. SFEE stands for?

— Steady Flow Energy Equation.

24. Turbine work (SFEE)?

— $h_1 - h_2$.

25. Nozzle exit velocity?

— $\sqrt{(2\Delta h)}$.

26. SFEE for boiler?

— $q = h_2 - h_1$.

27. 1 cal = ?

— 4.186 J.

28. Isothermal work?

— $p_1V_1 \ln(V_2/V_1)$.

29. Adiabatic work?

— $(p_1V_1 - p_2V_2)/(\gamma - 1)$.

30. Process representable on p–V?

— Quasi-static.

Practice MCQs — Basic Concepts & First Law of Thermodynamics

Q1. A system with energy but no mass transfer is:

(a) Open
(b) Closed
(c) Isolated
(d) Semi-open

✔ **Answer: (b) Closed**

Explanation: Closed system.

Q2. Which is an intensive property?

(a) Volume
(b) Energy
(c) Pressure
(d) Enthalpy

✔ **Answer: (c) Pressure**

Explanation: Independent of mass.

Q3. Heat and work are:

(a) Properties
(b) Point functions
(c) Path functions
(d) Intensive properties

✔ **Answer: (c) Path functions**

Explanation: Depend on the path.

Q4. First law for a closed system:

(a) $Q = \Delta U + W$
(b) $Q = \Delta H + W$
(c) $Q = W$
(d) $Q = \Delta U - W$

✔ **Answer: (a) $Q = \Delta U + W$**

Explanation: Energy conservation.

Q5. Work in a constant volume process:

(a) Maximum
(b) Zero
(c) p ΔV
(d) Infinite

✔ **Answer: (b) Zero**

Explanation: dV = 0.

Q6. In a throttling process, constant is:

(a) Temperature
(b) Pressure
(c) Enthalpy
(d) Entropy

✔ **Answer: (c) Enthalpy**

Explanation: $h_1 = h_2$.

Q7. Thermometry is based on:

(a) Zeroth law
(b) First law
(c) Second law
(d) Third law

✔ **Answer: (a) Zeroth law**

Explanation: Thermal equilibrium.

Q8. Nozzle enthalpy drop 125 kJ/kg — exit velocity (inlet negligible):

(a) 250 m/s
(b) 354 m/s
(c) 500 m/s
(d) 125 m/s

✔ **Answer: (c) 500 m/s**

Explanation: √(2 × 125000) = 500.

Q9. In free expansion:

(a) W = 0, Q = 0
(b) W = 0, Q ≠ 0
(c) W ≠ 0, Q = 0
(d) ΔU ≠ 0

✔ **Answer: (a) W = 0, Q = 0**

Explanation: Expansion into vacuum, ΔU = 0.

Q10. Value of γ for air:

(a) 1.0
(b) 1.33
(c) 1.4
(d) 1.67

✔ **Answer: (c) 1.4**

Explanation: c_p/c_v = 1.005/0.718.

Q11. PMM-I violates:

(a) Zeroth law
(b) First law
(c) Second law
(d) Third law

✔ **Answer: (b) First law**

Explanation: Energy conservation.

Q12. In an isothermal process of an ideal gas:

(a) $Q = 0$ (b) $W = 0$

(c) $\Delta U = 0$ (d) $\Delta H \neq 0$

✔ **Answer: (c) $\Delta U = 0$**

Explanation: U depends only on T.

Q13. Turbine work from SFEE (KE, PE, Q negligible):

(a) $h_1 + h_2$ (b) $h_1 - h_2$

(c) $h_2 - h_1$ (d) $u_1 - u_2$

✔ **Answer: (b) $h_1 - h_2$**

Explanation: Enthalpy drop.

Q14. Total energy of an isolated system:

(a) Increases (b) Decreases

(c) Remains constant (d) Becomes zero

✔ **Answer: (c) Remains constant**

Explanation: No exchange.

Q15. Heat supplied at constant pressure equals change in:

(a) Internal energy (b) Enthalpy

(c) Entropy (d) Work

✔ **Answer: (b) Enthalpy**

Explanation: $Q = \Delta H$.

SSC JE / UPSSSC JE Previous Year Questions — Basic Concepts & First Law of Thermodynamics

Q1. The steady flow energy equation is based on:

SSC JE (Previous Year)

(a) Mass conservation (b) Energy conservation

(c) Momentum conservation (d) Second law

✔ **Answer: (b) Energy conservation**

Explanation: First law.

Q2. When a system completes a cycle:

SSC JE (Previous Year)

(a) $\oint \delta Q = \oint \delta W$ (b) $\oint \delta Q = 0$

(c) $\oint \delta W = 0$ (d) $\oint dU \neq 0$

✔ **Answer: (a) $\oint \delta Q = \oint \delta W$**

Explanation: First law for a cycle.

Q3. Which is not a property? SSC JE (Previous Year)

(a) Pressure
(b) Temperature
(c) Heat
(d) Enthalpy

✔ **Answer: (c) Heat**

Explanation: Heat is a path function.

Q4. In an adiabatic process: SSC JE (Previous Year)

(a) No heat transfer
(b) Constant temperature
(c) Constant pressure
(d) Constant volume

✔ **Answer: (a) No heat transfer**

Explanation: Q = 0.

Q5. Throttling is a ____ process: SSC JE (Previous Year)

(a) Reversible
(b) Irreversible isenthalpic
(c) Isothermal
(d) Isentropic

✔ **Answer: (b) Irreversible isenthalpic**

Explanation: h constant, irreversible.

Q6. Example of an open system: SSC JE (Previous Year)

(a) Gas in a closed cylinder
(b) Steam turbine
(c) Thermos flask
(d) Sealed tank

✔ **Answer: (b) Steam turbine**

Explanation: Mass flows through.

UPSSSC ALERT

Chapter 3: Second Law of Thermodynamics

Second Law of Thermodynamics – Basic Concept

The first law only accounts for energy; it does not give direction. The **second law** states that heat flows by itself only from hot to cold and that heat cannot be fully converted into work. The limits of heat engines, heat pumps and refrigerators come from this law.

Understand in Detail — Second Law of Thermodynamics

3.1 Why was the second law needed?

The first law says the total energy account will balance, but it does **not** say in which **direction** a process will go. Hot tea cools by itself, but cold tea does not heat up by itself by taking heat from the room — though energy would be conserved in both! Similarly work can be fully converted to heat (when braking), but heat cannot be fully converted to work. **The second law tells the direction of processes and the 'quality' of energy.**

Thermal reservoir — a source so large that its temperature does not change on giving/taking heat (ocean, atmosphere, furnace). The one giving heat = **source**, the one receiving heat = **sink**.

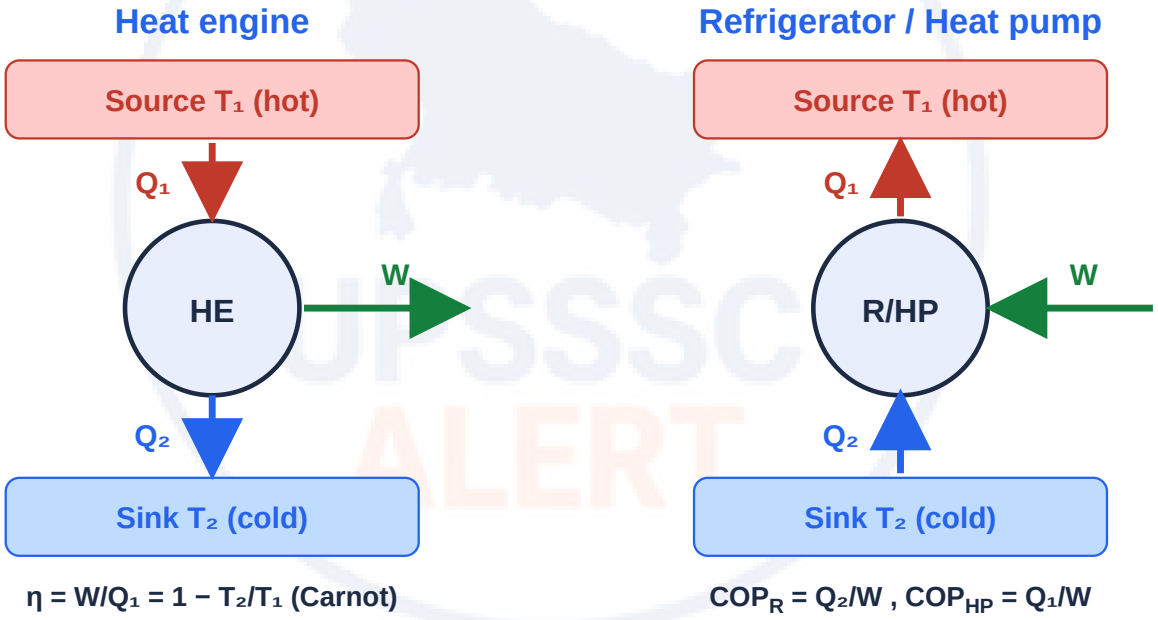


Fig 4.3: Heat engine and refrigerator/heat pump

3.2 Kelvin-Planck and Clausius statements — both say the same thing

Kelvin-Planck statement (for heat engines): it is impossible to build a cyclic device that takes heat from **only one** reservoir and converts it completely into work. That is, every heat engine must reject some heat to a **cold sink → no engine can be 100% efficient**. Such an imaginary machine = **PMM-2**.

Clausius statement (for refrigerators): it is impossible to build a device that transfers heat from a colder to a hotter body without external work. That is, a fridge must need electricity.

The two are equivalent: suppose a fridge violating Clausius existed (moving heat from cold to hot without work). Couple it with an ordinary engine that rejects the same heat to the sink — the sink's net exchange becomes zero and the combined machine would take heat only from the source and give complete work — violating Kelvin-Planck too. So violating one means violating the other.

Derivation: Engine, refrigerator and heat pump — measures of performance

1. **Heat engine:** $\eta = W/Q_1 = 1 - Q_2/Q_1$ (Q_1 = taken from source, Q_2 = given to sink)
2. **Refrigerator:** $COP_R = Q_2/W = Q_2/(Q_1 - Q_2)$
3. **Heat pump:** $COP_{HP} = Q_1/W = Q_1/(Q_1 - Q_2)$; $\mathbf{COP_{HP} = COP_R + 1}$

3.3 Carnot cycle — the limit of the most efficient engine

A **reversible process** is one which, when reversed, returns both the system and the surroundings exactly to their original states, leaving no trace. Real processes are **irreversible** — causes: friction, heat transfer across a finite temperature difference, free expansion, mixing, electrical resistance. A reversible process is an ideal limit.

The **Carnot cycle** (Sadi Carnot, 1824) — a completely reversible cycle with four processes: **1→2** isothermal expansion — heat Q_1 taken from the source (T_1); **2→3** adiabatic expansion — temperature T_1 to T_2; **3→4** isothermal compression — Q_2 rejected to the sink (T_2); **4→1** adiabatic compression — temperature T_2 to T_1. On a T–s diagram it is a **rectangle**.

Derivation: Carnot efficiency

1. On the T–s diagram: $Q_1 = T_1(s_2 - s_1)$, $Q_2 = T_2(s_2 - s_1)$ (sides of the rectangle)
2. $\eta = 1 - Q_2/Q_1 = \mathbf{1 - T_2/T_1}$ (temperatures in kelvin!)
3. $COP_R = T_2/(T_1 - T_2)$; $COP_{HP} = T_1/(T_1 - T_2)$

Carnot theorem: (1) no engine working between two given reservoirs can be more efficient than a reversible (Carnot) engine; (2) all reversible engines between the same reservoirs have equal efficiency, whatever the working substance. **Practical meaning:** to raise efficiency, raise the source temperature and lower the sink temperature. The Carnot cycle is not practical, because the isothermal process must be very slow and the adiabatic very fast — impossible in one machine.

Example: An inventor claims his engine gives 75% efficiency between 1000 K and 300 K. Check the claim.

Solution:

1. $\eta_{Carnot} = 1 - 300/1000 = \mathbf{70\%}$
2. Claim 75% > 70% — **impossible**, since no engine can be more efficient than Carnot

3.4 Entropy — the measure of 'disorder'

Clausius inequality: for any cycle $\oint \delta Q/T \le 0$ — equal (= 0) for a reversible cycle, less for an irreversible one. Since the cyclic integral of δQ/T is zero for a reversible cycle, it is a **property** — called **entropy (S)**: $\mathbf{dS = \delta Q_{rev}/T}$, unit kJ/K.

Simple meaning: entropy is a measure of molecular disorder (randomness). Ice (ordered) → water → steam (most disordered) — entropy increases. **Principle of increase of entropy:** the entropy of an isolated system (or the universe) never decreases — constant in a reversible process, increasing in an irreversible one: $\mathbf{\Delta S_{universe} \ge 0}$. This is the most general form of the second law.

Entropy change of an ideal gas: $\Delta S = m[c_v \ln(T_2/T_1) + R \ln(V_2/V_1)] = m[c_p \ln(T_2/T_1) - R \ln(p_2/p_1)]$. Isothermal: $\Delta S = mR \ln(V_2/V_1)$; **reversible adiabatic: ΔS = 0 → called isentropic**. On heating water $\Delta S = m\, c \ln(T_2/T_1)$; on melting/boiling ΔS = m L/T.

> **Example:** 1 kg of ice at 0°C melts into water (L = 335 kJ/kg). Entropy change?
> **Solution:**
> 1. $\Delta S = mL/T = 1 \times 335/273 =$ **1.227 kJ/K** — positive, because disorder increased

> **Note: Available energy (exergy):** the part of heat that can be converted into maximum work relative to the environment (T_0) = $Q(1 - T_0/T)$; the rest is **unavailable energy** = $T_0\, \Delta S$. Irreversibility destroys available energy — so the 'energy crisis' is really an 'available-energy crisis'.

1. Key Points

- **Thermal reservoir:** a body of such large heat capacity that its temperature does not change on heat exchange. The one supplying heat is the **source**, the one receiving it is the **sink**.
- **Kelvin–Planck statement:** no cyclic engine can receive heat from a single reservoir and convert it completely into work — a 100% efficient heat engine (PMM-II) is impossible.
- **Clausius statement:** heat cannot flow by itself from a colder to a hotter body without external work — a refrigerator needs work input.
- **The two statements are equivalent:** violating one implies violating the other.
- **Heat engine (HE):** takes Q_1 from the source, does work W and rejects Q_2 to the sink; $\eta = W/Q_1 = 1 - Q_2/Q_1$.
- **Refrigerator:** removes Q_2 from the cold space and rejects Q_1 to the hot; $COP_R = Q_2/W = T_2/(T_1 - T_2)$ (Carnot).
- **Heat pump:** aims to supply Q_1 to the hot space; $COP_{HP} = Q_1/W = T_1/(T_1 - T_2)$; and $COP_{HP} = COP_R + 1$.
- **Carnot cycle:** two isothermal and two isentropic (reversible adiabatic) processes; $\eta_{Carnot} = 1 - T_2/T_1$ (T in kelvin). It is the maximum possible efficiency between two given reservoirs (Carnot's theorem).
- **Entropy:** $dS = \delta Q_{rev}/T$; the entropy of an isolated system never decreases ($\Delta S \ge 0$). A reversible adiabatic process is isentropic.

- **Causes of irreversibility:** friction, heat transfer across a finite temperature difference, free expansion, mixing and inelastic deformation.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Heat engine efficiency	**$\eta = W/Q_1 = 1 - Q_2/Q_1$**
Carnot efficiency	**$\eta = 1 - T_2/T_1$**
Refrigerator COP	**$COP_R = Q_2/W = T_2/(T_1 - T_2)$**
Heat pump COP	**$COP_{HP} = Q_1/W = T_1/(T_1 - T_2)$**
Relation	**$COP_{HP} = COP_R + 1$**
Energy balance	**$Q_1 = W + Q_2$**
Entropy	**$dS = \delta Q_{rev}/T$**
Clausius inequality	**$\oint \delta Q/T \leq 0$**
1 ton of refrigeration	**≈ 3.5 kW (210 kJ/min)**

3. Detailed Explanation

- **Four processes of the Carnot cycle:** (1–2) isothermal expansion at T_1 (Q_1 added), (2–3) isentropic expansion, (3–4) isothermal compression at T_2 (Q_2 rejected), (4–1) isentropic compression. On a T–s diagram it is a rectangle.
- **Corollaries of Carnot's theorem:** (i) no engine between the same reservoirs is more efficient than a reversible one; (ii) all reversible engines have the same efficiency — independent of the working substance.
- **Practical limitation:** the Carnot cycle is not practical, as isothermal processes must be infinitely slow and adiabatic ones infinitely fast; but it is the standard for comparison.
- **Raising efficiency:** increase T_1 or decrease T_2; lowering T_2 is more effective, but the sink is usually the atmosphere, which limits it.
- **Use of a heat pump:** heating rooms in winter; its COP is always greater than 1, so it is cheaper than an electric heater.

4. Comparison Table — Heat Engine, Refrigerator and Heat Pump

Basis	Heat engine	Refrigerator	Heat pump
Purpose	Produce work	Remove heat from cold space	Supply heat to hot space
Work	Output	Input	Input
Performance measure	$\eta = W/Q_1$	$COP = Q_2/W$	$COP = Q_1/W$
Value	Always < 1	> 1 or < 1	Always > 1

5. Solved Numerical Problems

Example 1. A Carnot engine works between 527°C and 27°C. Find its efficiency.

Solution:

1. $T_1 = 800$ K, $T_2 = 300$ K
2. $\eta = 1 - 300/800 =$ **62.5%**

Example 2. An engine receives 1000 kJ and produces 400 kJ of work. Find heat rejected and efficiency.

Solution:

1. $Q_2 = 1000 - 400 =$ **600 kJ**
2. $\eta = 400/1000 =$ **40%**

Example 3. A Carnot refrigerator works between −13°C and 27°C. Find COP_R and COP_{HP}.

Solution:

1. $T_2 = 260$ K, $T_1 = 300$ K
2. $COP_R = 260/40 =$ **6.5**
3. $COP_{HP} = 6.5 + 1 =$ **7.5**

Example 4. A refrigerator removes 5 kW of heat with COP = 4. Find power input and heat rejected.

Solution:

1. $W = Q_2/COP = 5/4 =$ **1.25 kW**
2. $Q_1 = 5 + 1.25 =$ **6.25 kW**

Example 5. An inventor claims an engine working between 400 K and 300 K with 30% efficiency. Is it possible?

Solution:

1. $\eta_{Carnot} = 1 - 300/400 = 25\%$
2. Claim (30%) > Carnot (25%), so **impossible**

6. Frequently Asked Facts

- Carnot efficiency $1 - T_2/T_1$, temperatures in kelvin.
- $COP_{HP} = COP_R + 1$.
- Kelvin–Planck — heat engine; Clausius — refrigerator.
- PMM-II violates the second law.

7. Note (Confusing Facts)

- Use kelvin, not °C, in the Carnot formula — the most common mistake.
- COP can exceed 1; efficiency cannot.
- A reversible engine's efficiency does not depend on the working substance.

Remember: One line: $\eta_{max} = 1 - T_2/T_1$; $COP_R = T_2/(T_1 - T_2)$. Assertion–Reason: (A) A 100% efficient heat engine is impossible. (R) Some heat must be rejected to a sink (Kelvin–Planck). — Both true; R explains A.

30 Most Important One-Liner Facts — Second Law of Thermodynamics

1. Reservoir supplying heat?

— Source.

2. Reservoir receiving heat?

— Sink.

3. Statement about heat engine?

— Kelvin–Planck.

4. Statement about refrigerator?

— Clausius.

5. PMM-II violates?

— Second law.

6. Heat engine efficiency?

— W/Q_1.

7. Carnot efficiency?

— $1 - T_2/T_1$.

8. Processes of Carnot cycle?

— 2 isothermal + 2 isentropic.

9. Shape of Carnot cycle on T–s?

— Rectangle.

10. Refrigerator COP?

— Q_2/W.

11. Heat pump COP?

— Q_1/W.

12. $COP_{HP} - COP_R$?

— 1.

13. Carnot refrigerator COP?

— $T_2/(T_1 - T_2)$.

14. Heat pump COP is always?

— Greater than 1.

15. Engine efficiency is always?

— Less than 1.

16. Max efficiency between two reservoirs?

— Carnot (reversible).

17. Reversible engine efficiency depends on?

— Only reservoir temperatures.

18. Definition of entropy?

— $dS = \delta Q_{rev}/T$.

19. Entropy of isolated system?

— Never decreases.

20. Reversible adiabatic process?

— Isentropic.

21. Unit of entropy?

— kJ/K.

22. Clausius inequality?

— $\oint \delta Q/T \leq 0$.

23. A cause of irreversibility?

— Friction.

24. 1 TR approximately?

— 3.5 kW.

25. Energy balance (HE)?

— $Q_1 = W + Q_2$.

26. More effective way to raise η?

— Lower T_2.

27. Device to heat room in winter?

— Heat pump.

28. Heat flows by itself?

— Hot to cold.

29. Why Carnot impractical?

— Needs infinitely slow/fast processes.

30. Feature of thermal reservoir?

— Temperature stays constant.

Practice MCQs — Second Law of Thermodynamics

Q1. Efficiency of a Carnot engine:

(a) $1 - T_1/T_2$ (b) $1 - T_2/T_1$

(c) T_2/T_1 (d) $T_1/(T_1 - T_2)$

✔ **Answer: (b) $1 - T_2/T_1$**

Explanation: T in kelvin.

Q2. Kelvin–Planck statement relates to:

(a) Refrigerator (b) Heat engine

(c) Heat pump (d) Compressor

✔ **Answer: (b) Heat engine**

Explanation: 100% efficient engine impossible.

Q3. Relation of COP_{HP} and COP_R:

(a) $COP_{HP} = COP_R - 1$ (b) $COP_{HP} = COP_R + 1$

(c) $COP_{HP} = 2COP_R$ (d) $COP_{HP} = 1/COP_R$

✔ **Answer: (b) $COP_{HP} = COP_R + 1$**

Explanation: $Q_1 = Q_2 + W$.

Q4. Carnot efficiency between 327°C and 27°C:

(a) 50% (b) 91.7%

(c) 8.3% (d) 25%

✔ **Answer: (a) 50%**

Explanation: $1 - 300/600 = 0.5$.

Q5. PMM-II violates:

(a) First law (b) Second law

(c) Zeroth law (d) None

✔ **Answer: (b) Second law**

Explanation: Complete conversion from one reservoir.

Q6. The Carnot cycle consists of:

(a) 2 isothermal + 2 isobaric (b) 2 isothermal + 2 isentropic

(c) 2 isochoric + 2 isentropic (d) 2 isobaric + 2 isentropic

✔ **Answer: (b) 2 isothermal + 2 isentropic**

Explanation: Carnot.

Q7. COP of a Carnot refrigerator between −3°C and 27°C:

(a) 9
(b) 10
(c) 0.9
(d) 11

✔ **Answer: (a) 9**

Explanation: 270/30 = 9.

Q8. COP of a heat pump is:

(a) Always less than 1
(b) Always more than 1
(c) Equal to 1
(d) Zero

✔ **Answer: (b) Always more than 1**

Explanation: $COP_R + 1$.

Q9. Heat cannot flow from cold to hot without work — this is:

(a) Kelvin–Planck
(b) Clausius
(c) Carnot
(d) Joule

✔ **Answer: (b) Clausius**

Explanation: Clausius statement.

Q10. Entropy of an isolated system in an irreversible process:

(a) Decreases
(b) Increases
(c) Constant
(d) Zero

✔ **Answer: (b) Increases**

Explanation: $\Delta S > 0$.

Q11. Efficiency of an engine taking 1000 kJ and giving 300 kJ work:

(a) 30%
(b) 70%
(c) 43%
(d) 3.3%

✔ **Answer: (a) 30%**

Explanation: 300/1000.

Q12. Efficiency of a reversible engine depends on:

(a) Working substance
(b) Reservoir temperatures
(c) Engine size
(d) Speed

✔ **Answer: (b) Reservoir temperatures**

Explanation: Carnot's theorem.

Q13. 1 ton of refrigeration is approximately:

(a) 1 kW
(b) 3.5 kW
(c) 10 kW
(d) 746 W

✔ **Answer: (b) 3.5 kW**

Explanation: 210 kJ/min ≈ 3.5 kW.

Q14. On a T–s diagram, the Carnot cycle appears as:

(a) Circle
(b) Rectangle
(c) Triangle
(d) Parabola

✔ **Answer: (b) Rectangle**

Explanation: Two horizontal and two vertical lines.

Q15. More effective way to increase heat engine efficiency:

(a) Increase T_1
(b) Decrease T_2
(c) Both equal
(d) Neither

✔ **Answer: (b) Decrease T_2**

Explanation: Efficiency is more sensitive to T_2.

SSC JE / UPSSSC JE Previous Year Questions — Second Law of Thermodynamics

Q1. COP of a heat pump compared to that of a refrigerator is:

SSC JE (Previous Year)

(a) Greater by 1
(b) Less by 1
(c) Equal
(d) Double

✔ **Answer: (a) Greater by 1**

Explanation: $COP_{HP} = COP_R + 1$.

Q2. Efficiency of the Carnot cycle depends on:

SSC JE (Previous Year)

(a) Working substance
(b) Only source and sink temperatures
(c) Pressure
(d) Volume

✔ **Answer: (b) Only source and sink temperatures**

Explanation: $\eta = 1 - T_2/T_1$.

Q3. By the second law, the efficiency of a heat engine is:

SSC JE (Previous Year)

(a) Can be 100%
(b) Always less than 100%
(c) More than 100%
(d) Zero

✔ **Answer: (b) Always less than 100%**

Explanation: Kelvin–Planck.

Q4. Entropy is associated with:

SSC JE (Previous Year)

(a) Zeroth law
(b) First law
(c) Second law
(d) Newton's law

✔ **Answer: (c) Second law**

Explanation: Second law.

Q5. A refrigerator works on: SSC JE (Previous Year)

(a) Carnot cycle
(b) Reversed Carnot cycle
(c) Otto cycle
(d) Diesel cycle

✔ **Answer: (b) Reversed Carnot cycle**

Explanation: Ideally reversed Carnot.

Q6. Maximum efficiency of an engine between 27°C and 327°C: SSC JE (Previous Year)

(a) 50%
(b) 92%
(c) 8%
(d) 100%

✔ **Answer: (a) 50%**

Explanation: 1 − 300/600.

UPSSSC ALERT